WHEELTEC

WHEELTEC H30 Series Inertial Navigation User Manual

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Release Notes：

Version	dates	Description of content
V1.0	2024/01/02	First release
V2.0	2024/05/22	Added ROS2_SDK related content

Website：www.wheeltec.net

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1. Overview

1. 1 Brief description of WHEELTEC H30

The WHEELTEC H30 is a highly accurate attitude sensor that measures the carrier's 3D attitude angle, acceleration, angular velocity and magnetic field strength information.

WHEELTEC H30 integrates industrial-grade, highly reliable 3-axis MEMS gyroscope, 3-axis MEMS accelerometer and 3-axis magnetic sensor. With embedded YFusion® high-performance attitude fusion algorithm, high-precision sensor error compensation algorithms, and rigorous factory-tested calibration,WHEELTEC H30 is able to achieve 0.1° roll and pitch angle measurement accuracy, 0.5° no-reference heading angle, 1° magnetic reference heading angle measurement accuracy.

WHEELTEC H30 supports UART and I2C interface communication, as well as synchronization signal input, and is reserved with 485 and UART switching switches foreasy customer integration.

Figure 1.1.1 WHEELTEC H30 Series Inertial Guidance Product Map

The system uses Yesense private protocolfor communication, please refer to the document “4. YIS Communication Protocol” for the related data format and protocol, and its core algorithm schematic is shown in Figure 1.1.2.

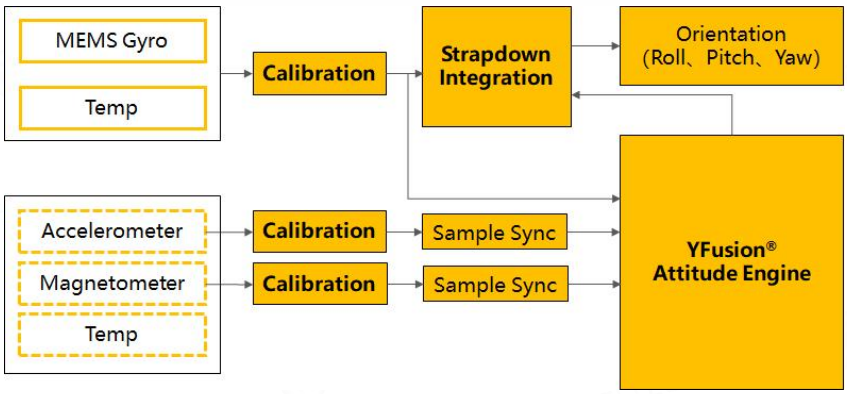


Figure 1.1.2 System Block Diagram

1. 2 Description of WHEELTEC H30 parameters

TheWHEELTEC H30 inertial navigation is used in mobile robots, drones, lawnmowers and so on. It consists of a three-axis gyroscope, three-axis accelerometer, three-axis magnetometer, and a thermometer as part of the nine-axis Attitude Sensor family. The WHEELTEC H30 hardware sensor parameters are shown in the following figure.

IMU	accelerometer	gyros	magnetometer
range	±16g	±2000°/s	±49Gauss
Zero-bias instability	0.035mg	4°/h	/
nonlinearity	±0.5%FS	±0.05%FS	±0.2%FS
noise intensity	60μg/√Hz	0.005°/s/√Hz	3mGauss
Bandwidth (-3db)	42Hz	49Hz	/
zero offset	±10mg	±1°/s	/
zero-point temperature drift	±20mg	±1°/s	/

Table 1.2.1 WHEELTEC H30 sensor parameters

The WHEELTEC H30 navigation performance parameters are shown below.

parameters		typical value	note
roll angle	precision	0.1°	1σ RMS,25℃
	range	±180°	
tilt angle	precision	0.1°	1σ RMS,25℃
	range	±90°	
No reference heading angle	precision	0.5°	1σ RMS,25℃
	range	±180°	
magnetic reference heading angle	precision	1°	
	range	±180°	
angular resolution		0.01°	

Table 1.2.2 WHEELTEC H30 Navigation Performance Parameters

2. Wiring instructions and testing

2. 1 WHEELTEC H30 base plate pin distribution

The WHEELTEC H30 pin distribution and corresponding functions are described in the chart below.Type-Cserial port outputis used by default.

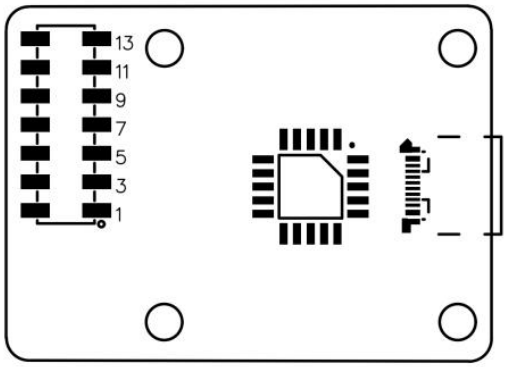


Figure 2.1.1 WHEELTEC H30 pin distribution

The functions of the corresponding pins in the diagram are described in Table 2.1.1, where the serial port baud rate defaults to 406800 bps.

Table 2.1.1 EVB Pin Descriptions

name	Pin number	I/O		note
VBUS	1	I	Power Input	4.5~5.2V DC
VBUS	2	I	Power Input	
GND	3		Power Ground	
GND	4		Power Ground	
RX_URAT	5	I	UART receive, TTL level	3.3V
TX_URAT	7	O	UART transmit, TTL level	3.3V
SYNC	6	I	Synchronization Input Pins	
485A	8		485A+	
485B	10		485B-	
SDA	9	O	I2C Data Pin	Internal pull-up
SCL	11	I	I2C Clock pin	Internal pull-up
INT	12	O	Synchronous output pin	Internal drop-down
GND	13		Power Ground	
3.3V	14	O	3.3V DC power supply	Power output only

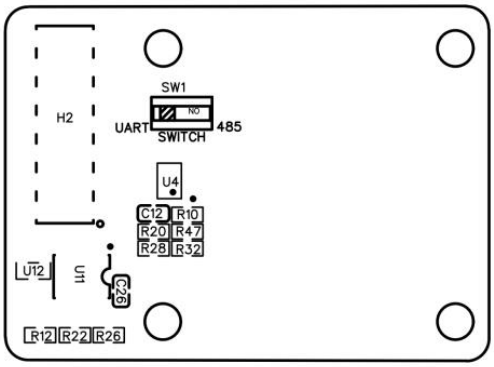


Figure 2.1.2 WHEELTEC H30 Pin distribution diagram

The part indicated in the above figure is the part of the circuit device that needs to be soldered when the module uses RS485 and UART dip switches, which are not included in the default circuit. If you need to use the RS485 circuit, you can solder the corresponding components according to the schematic diagram.

2. 2 WHEELTEC H30 mini Pin distribution

The pin distribution and corresponding functions of the WHEELTEC H30 mini version are described in the chart below.Type-Cserial outputs used bydefault, where the serial baud rate is 406800 bps by default.

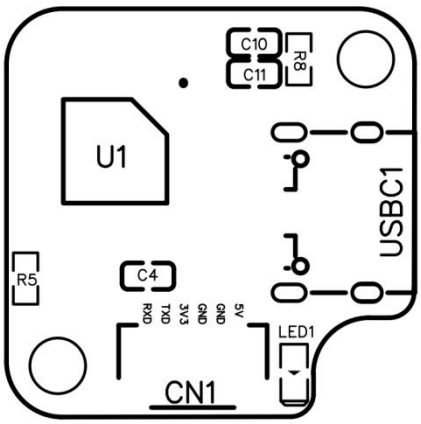


Figure 2.2.1 WHEELTEC H30 mini Pin distribution diagram

name	I/O	
5V	I	Power Input ,4.5~5.2V DC
GND		Power Ground
GND		Power Ground
3V3	O	Power Input
TX_URAT	O	UART transmit, TTL level
RX_URAT	I	UART receive, TTL level

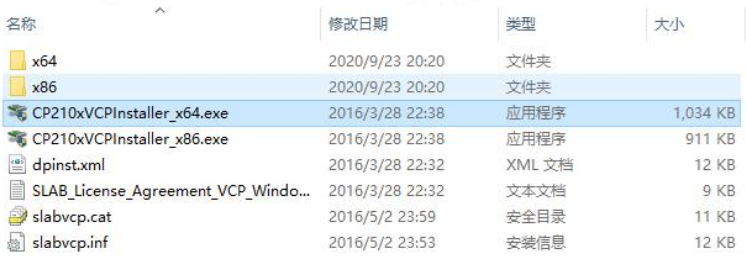
Table 2.2.1 WHEELTEC H30 minipin description

2. 3 Driver Installation

① Installation of drivers on Windows system

Windows systems require a driver torecognize WHEELTEC H30 series inertial navigation.

Open the “4.Software Tools-> CP2102 usb Driver.zip” in the attached packet, unzip it locally, select the corresponding file according to your computer's architecture, and double-click to run it.



名称	修改日期	类型	大小
x64	2020/9/23 20:20	文件夹	
x86	2020/9/23 20:20	文件夹	
CP210xVCPInstaller_x64.exe	2016/3/28 22:38	应用程序	1,034 KB
CP210xVCPInstaller_x86.exe	2016/3/28 22:38	应用程序	911 KB
dpinst.xml	2016/3/28 22:32	XML 文档	12 KB
SLAB_License_Agreement_VCP_Windo...	2016/3/28 22:32	文本文档	9 KB
slabvcp.cat	2016/5/2 23:59	安全目录	11 KB
slabvcp.inf	2016/5/2 23:53	安装信息	12 KB

Figure 2.3.1CP2102 driver installation

Follow the prompts and click next to complete the installation, as shown in the figure.

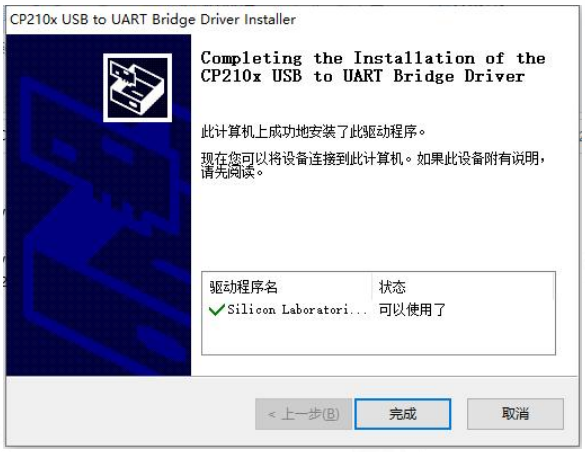


Figure 2.3.2 CP2102 Driver Installation

② **Installation of the driver on the Linux system**

Installation of the CP2102 driver is generally not required in Linux.

2. 4 Wiring Instructions

The WHEELTEC H30 series uses a 5V power supply and a TYPE-C connector to provide power and data transfer to the system. If you want to use the pins to power the system and expand the system, please note that the maximum withstand value of each pin in the module is 5.5V, and use VBUS as the power input.

After powering up, thered LED on the system board will be a constant state, and the blue LED on the system board will light up when there is a data transfer from the WHEELTEC H30.

Figure 2.4.1 WHEELTEC H30 without shell

2. 5 Power-up Detection

1. device connection

After connecting the IMU module to the computer via TYPE-C to USB, open the “WHEELTEC_H30 Inertial navigation Product Information\3.SoftwareTools\1.YIS

Manager uplink",double click Yesense-V2.8.5.exe to open the uplink tool.

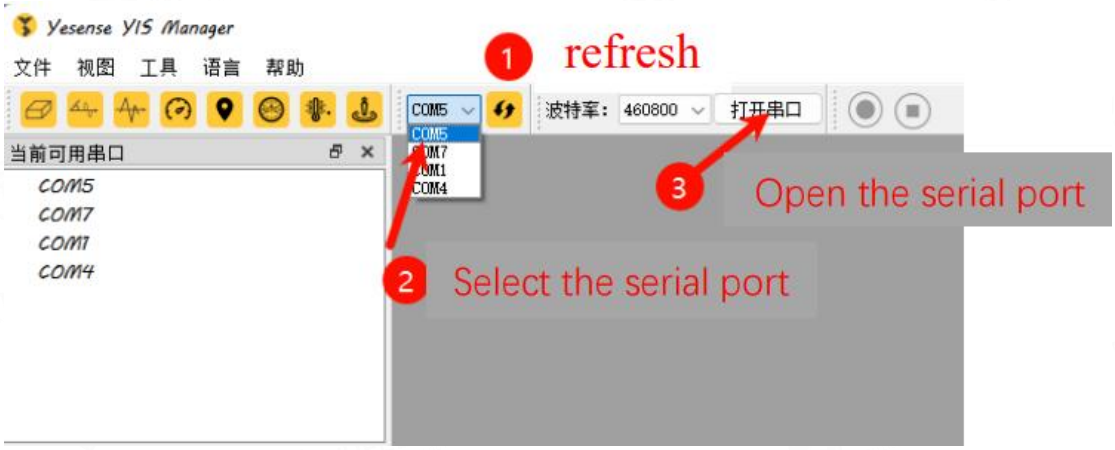


Figure 2.5.1 Upper computer device connection interface

1. First click on Refresh in the menu bar
2. Select the serial port corresponding to the module (for example: com5) and the baud rate of the serial output (default baud rate: 460800)
3. Open the serial port
4. Check product information and product status in the left column

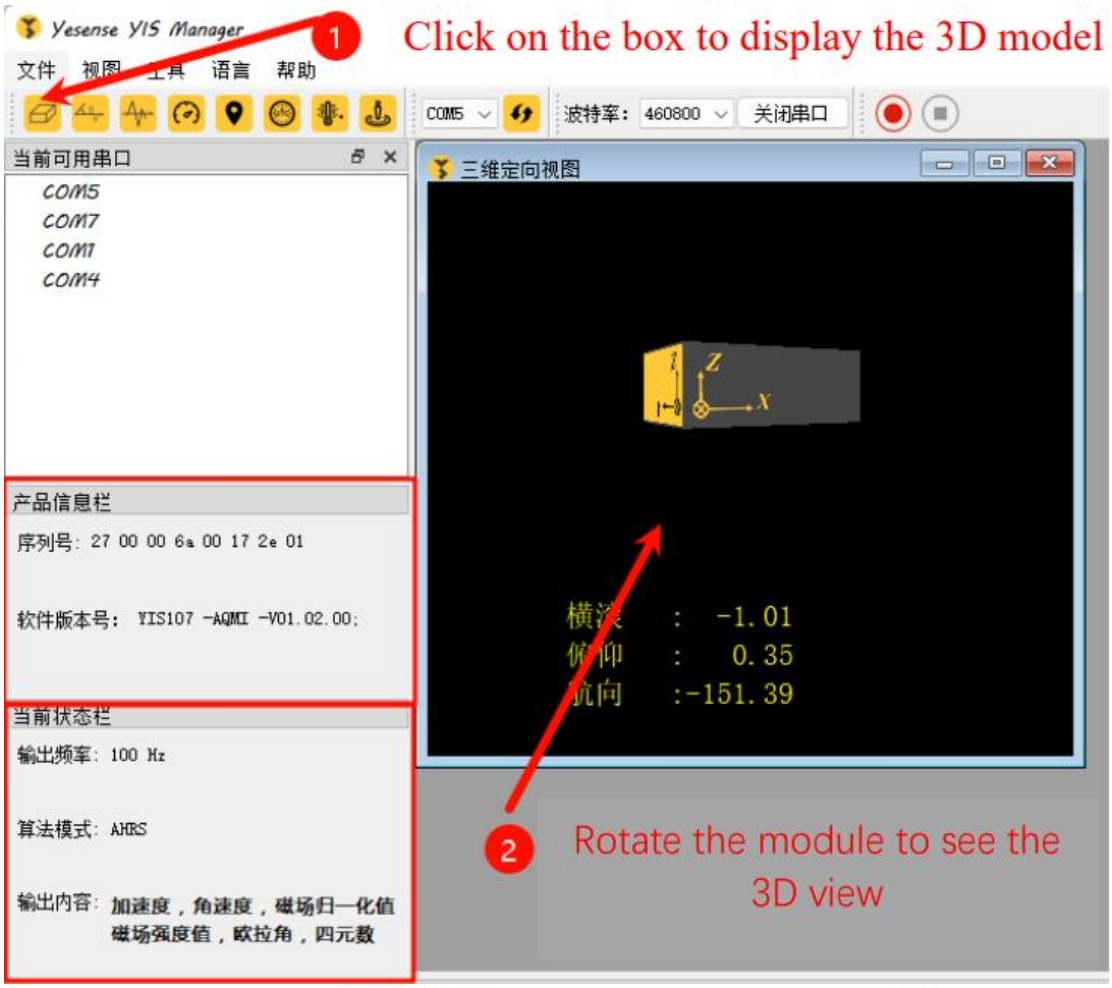


Figure 2.5.2 Upper computer device information and position display interface

2. OrientationDetection：

After clicking【3D Display】in the upper left corner, rotate the WHEELTEC H30 module, and the accuracy of the WHEELTEC H30 orientation can be verified according to the change of the module position in the【3D Orientation View】interface.

Figure 2.5.3 Upper IMU raw data presentation

3. View data：

Click on【Attitude Data】and【Raw Data】in the upper left cornerto read the data from the WHEELTEC H30 module, and you can check or uncheck the corresponding individual data charts according to your needs.

2. 6 Direction Switching

In order to adapt to the needs of different application scenarios, WHEELTEC H30 supports three modes of switching: no reference heading, magnetometer initialized heading and magnetic reference heading.

Magnetometer initialized heading: use the heading angle of the magnetometer to do the initialization, the subsequent incremental part is done by the heading angle increment of the IMU, the final output heading angle is the relative magnetic heading.

Figure 2.6.1 Upper computer function configuration interface

1. Click [Tools]→[Related Configuration]→[Function Settings]in the menu bar respectively.
2. In the [Function Configuration] interface, select [AHRS], [VRU], [IMU] modes according to requirements

IMU：Inertial Measurement Unit that outputs acceleration and angular velocity. Does not output information such as attitude angle

VRU：IMU based on the built-in attitude solution algorithm, can output attitude information

AHRS：A modified algorithm based on VRU that solves for the full attitude of the object under test, including the absolute heading angle.

If a magnetometer initialization heading is required it can be written via a command from the serial port:

Figure 2.6.2 Upper computer command writing interface

1. Click [Tools]→[Command Write]in the menu bar respectively.
2. In the [Command Write] screen, output "59 53 4D 12 00 02 38 99 05".
3. Click [Send] to write it and it will take effect.

2. 7 Definition of Orientation

The WHEELTEC H30 carrier system uses the Front-Left-Up (FLU) coordinate system and the geographic coordinate system uses the East-North-Up (ENU)

coordinate system. The acceleration and gyro axes are shown below:

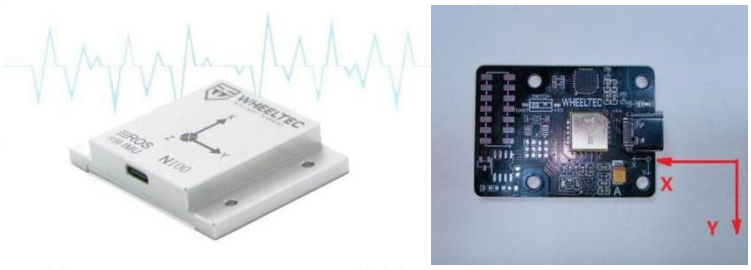


Figure 2.7.1 Product appearance

The Euler angle rotation order is East(X)-North(Y)-Sky(Z)-321(first turn the Z axis, then theY axis, and finally the X axis) rotation order. The specific definitions are as follows:

Rotation around Z-axis: Heading angle/Yaw/psi(ψ) Range: [-180°,180°]

Rotation around Y-axis:Pitch/Pitc/theta(θ) Range: [-90°,90°]

Rotation around X-axis direction: Roll angle/Roll/phi(ϕ)range: [-180°,180°]

3. Calibration and Leveling

3. 1 Gyro Zero Bias Calibration

When using WHEELTEC H30 series modules,conditions may occur where the output bit position data fluctuates greatly, requiring the system to undergoazero-bias calibration operation.

Figure 3.1.1 Upper computer toolbar interface

Figure 3.1.2 Upper computer gyro zero-bias estimation interface

1. Click [Tools]→[Reset & Calibration]→[Gyro Zero Bias]in the menu bar respectively.
2. Keep the device stationary and click [Measurement Estimation], the software starts to collect data and calculate the gyroscope zero deviation value.
3. After completing the acquisition, click [Write] to complete the gyro zero-bias calibration.

Note: Keep the device stationary during the estimation of the gyro zero bias value. If the device generates motion during the estimation process, it mayresult in abnormal device output performance.

3. 2 Attitude and Heading Angle reset

If you encounter a problem where the output pitch and roll angles are not zero, it means that there is a mounting error angle between the module and the mounting plane, andyou can perform an attitude angle reset.

Figure 3.2.1 Upper computer attitude reset and heading angle reset interface

3. 3 Magnetometer Calibration

The geomagnetic sensor has been ellipsoid calibrated before leaving the factory, but the magnetic sensor is easily interfered by the magnetic field of the external environment, and usually needs to be recalibrated by the customer after getting the product.Find a relatively empty place without interference,connect the module to the computer, open the upper computer.In front of the computer, please try to stay away from interference sources. Perform the following operation.

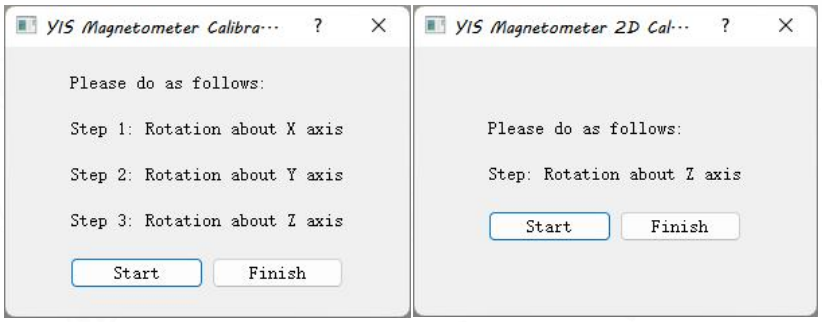


Figure 3.3.1 Upper unit magnetometer calibration interface

1. Install the YIS device and connect it to the PC.
2. Click [Tools]→[Reset & Calibrate]→[Magnetometer]in the menu bar, and select whether it is 3D or 2D calibration mode according to the specific application scenario.
3. Click [Start] to slowly move the module in all directions according to the 3D or 2D mode prompts.
4. Click [Finish] to generate 3D/2D calibration results.
5. Click the [Write to YIS] button, YIS Manager sends and writes the calibration parameters to the module, and the magnetic calibration process is completed.

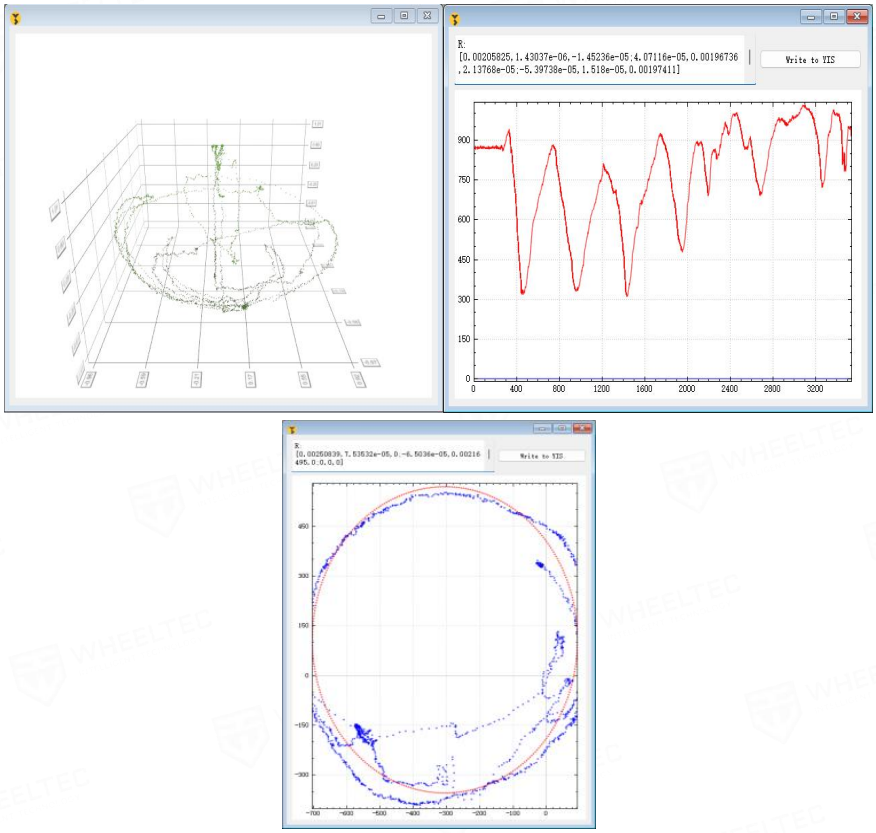


Figure 3.3.2 Upper magnetometer calibration interface

If you are not satisfied with the calibration accuracy, you can repeat the above steps for calibration by clicking the Start Calibration button again.

3. 4 Firmware upgrades

WHEELTEC H30 supports firmware upgrade function. Users can utilize this function by using the YESENSE YIS Manager upscaler.

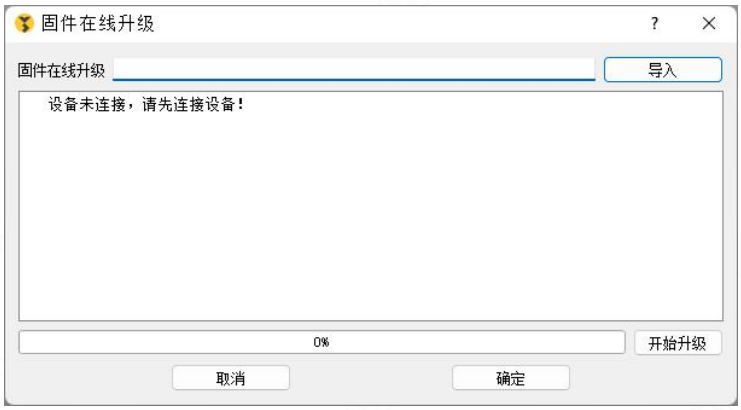


Figure 3.4.1 Upper computer firmware upgrade interface

1. Click [Help]→[Firmware Online Upgrade] in the menu bar respectively.
2. In the [Firmware Online Upgrade] interface, click [Import], select the binary .bin file
3. Click on [OK] to write to take effect

4. Common Configuration Commands for UART Interaction

4. 1 Communication baud rate configuration

The WHEELTEC H30 supports the following output baud rate configurations:

Table 4.1 Output baud rate configuration commands

Description		Directive（HEX）	Reference return value（HEX）
UART baud	9600 bps	59 53 02 0A 00 01 0D 27	none
	19200 bps	59 53 02 0A 00 06 12 2C	none
	38400 bps	59 53 02 0A 00 02 0E 28	none
	57600 bps	59 53 02 0A 00 07 13 2D	none
	115200 bps	59 53 02 0A 00 03 0F 29	none

	230400 bps	59 53 02 0A 00 09 15 2F	none
	460800 bps	59 53 02 0A 00 04 10 2A	none
	921600 bps	59 53 02 0A 00 05 11 2B	none

The baud rate setting needs to take into account the amount of data output in a single frame and the output frequency. Without changing the output content, if the baud rate is set too low, the data may not be completely output, resulting in unsuccessful configuration. When setting the output content and output frequency, if the communication parameters cannot meet the output requirements, the configuration will be unsuccessful. The minimum baud rate can be estimated according to the following formula:

1. Estimated single-frame data transmission time t = 1/f * 1000 - 1(ms)

f:Output frequency

2. Estimated baud rate baudrate = n* 10 *1000 /t

n: number of bytes of data output in a single frame.

baudrate:In order to ensure that commands can be received normally after modifying the baud rate, this value is recommended to be less than 80% of the baud rate that needs to be set.

Once the baud rate is set, connectthe WHEELTEC H30 modulewith the newly set baud rate.

4.2 Output Frequency Configuration

TheWHEELTEC H30 supports the following output frequency configurations:

Table 4.2.1 Output frequency configuration commands

Description		Directive（HEX）	Reference return value（HEX）	
Output frequency	1Hz	59 53 03 0A 00 01 0E 2B	successes	59 53 03 0A00 00 0D 2A
	2Hz	59 53 03 0A 00 02 0F 2C		
	5Hz	59 53 03 0A 00 03 10 2D		
	10Hz	59 53 03 0A 00 04 11 2E		
	20Hz	59 53 03 0A 00 05 12 2F		
	25Hz	59 53 03 0A 00 06 13	fail	59 53 03 0A 00 FF 0C 29

		30		
	50Hz	59 53 03 0A 00 07 14 31		
	100Hz	59 53 03 0A 00 08 15 32		
	200Hz	59 53 03 0A 00 09 16 33		

4. 3 Heading modes witching

The WHEELTEC H30 supports the following heading modes:

Table 4.3.1 Heading switching configuration commands

Description		Directive（HEX）	Reference return value（HEX）
Heading mode switching	No reference heading angle	59 53 4D 12 00 02 02 63 CF	59 53 4D 0A 00 00 57 52
	Magnetometer initialization heading	59 53 4D 12 00 02 38 99 05	
	magnetic reference heading	59 53 4D 12 00 02 01 62 CE	59 53 4D 0A 00 FF 56 51

5. Porting the ROS SDK

5. 1 CP2102 Fixed Serial Number

The serial port number corresponding to the peripheral is usually set by the master control according to the connection order of the device, such as CP2102, the device name is usually [ttyUSBx], which is usually changed, in order to avoid the situation of not finding the device, here you can solve the problem by fixing the serial port number for the USB device and creating an alias.

① **Modify the serial port number**

In Windows, you need to change the serial number of the CP2102 chip on the

WHEELTEC H30 series to 0003, connect the inertial module to the computer with a USB cable, and then modify and fix it through the windows software CP21xxCustomizationUtility, the operation is shown below:

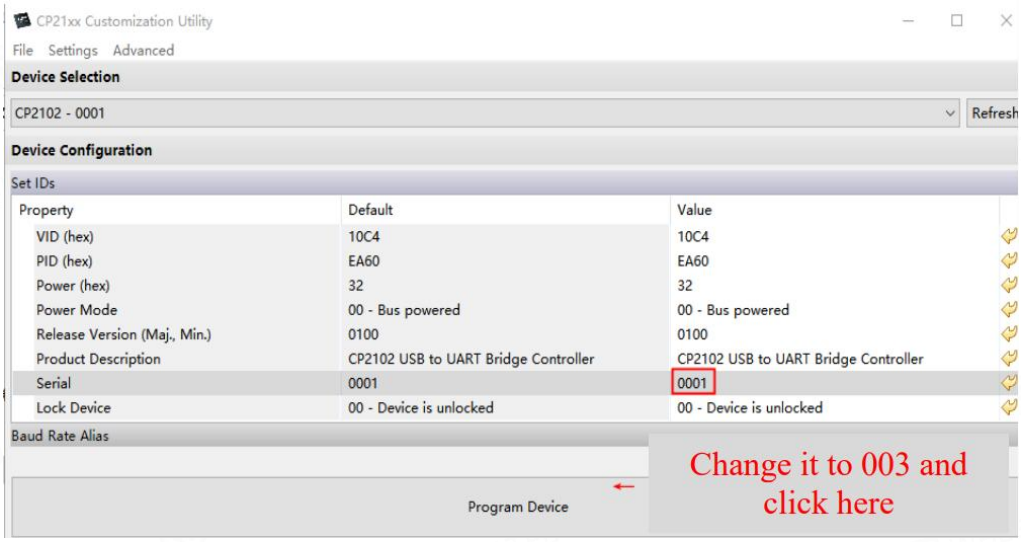


Figure 5.1.1 Modify the serial port number

② Create a device alias

The corresponding script file is stored in the wheeltec_udev.sh file in the [package \2.ROS_SDK\fdilink_ahrs_ROS1\fdilink_ahrs] folder. Run the following commands in the file directory to complete the alias creation.

```
Document empowerment: sudo chmod 777 wheeltec_udev.sh
Execute script: sudo ./wheeltec_udev.sh
```

③ Normal use

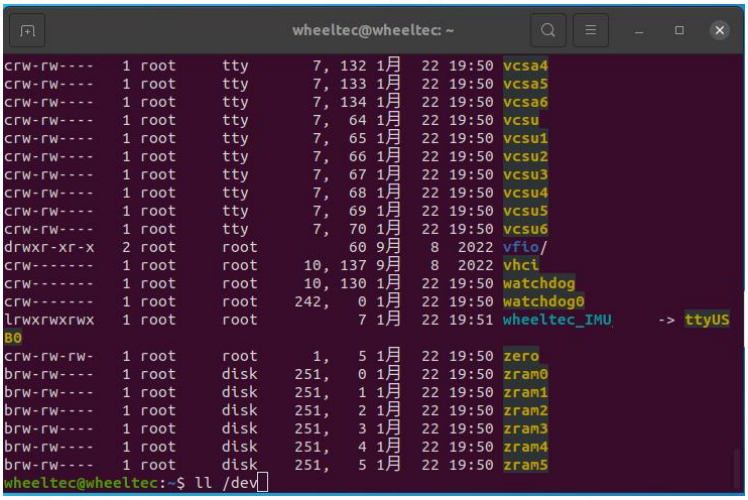


Figure 5.1.2 Modify the serial port number

Connect the WHEELTEC H30 module to the ROS master and run it on the

terminal: [ll /dev] to view the device

5. 2 CH9102 Fixed Serial Number

① Create a device alias

Here, we first confirm whether the module is connected. To prevent device ids from changing after multiple devices are connected to the system, you can bind devices to aliases using corresponding rules. Before binding the alias, we need to ensure that the device can be identified by the system. Connect the CH9102 device to the Ubuntu system and input the command "lsusb" to view the output. If the output item "Qinheng Electronics", it means that the driver has been connected to the system.

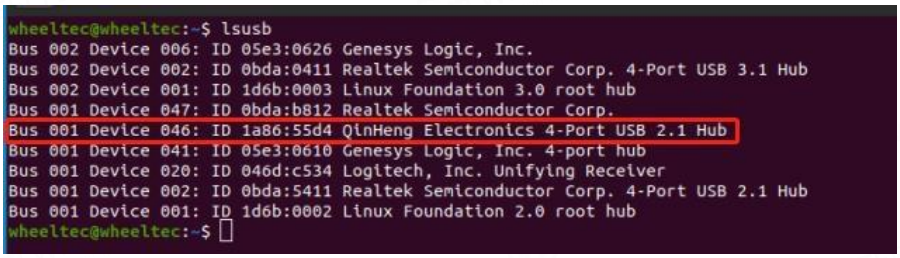


Figure 5.2.1 lsusb Command output

After the device is identified, you can create an alias for the device. Find the file [package \2.ROS_SDK\yesense_imu\wheeltec_udev.sh], which is the file for creating the alias rule of the CH9102 device, as shown in Figure 5.2.2.

In the preceding command, idVendor indicates the manufacturer ID of the CH9102 (default value: 1a86), and idProduct indicates the product ID of the CH9102 (default value: 55d4).

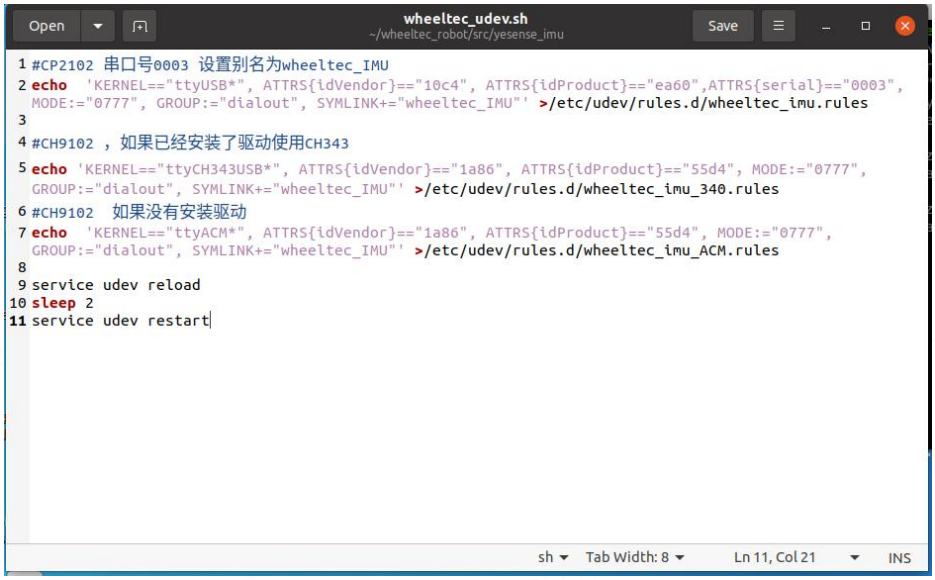


Figure 5.2.2 wheeltec_udev.sh

Copy the [wheeltec_udev.sh] file to the ubuntu system, and open the terminal in the folder where the file is located, then give the file 777 permission (If in doubt, check online: linux 777 permission), and then execute the file, grant permission and execute the file, the command is shown in the figure. After the script is executed, the system writes the device alias rule to /etc/udev/rules.d/. After the script is executed, remove and reinsert the device to make the alias rule take effect. Run the following command:

```
Document empowerment: sudo chmod 777 wheeltec_udev.sh
Execute script: sudo sh wheeltec_udev.sh
```

② **Normal use**

Connect the WHEELTEC H30 series module to the ROS master and run it at the terminal：

```
ll /dev
```

View device, display：wheeltec_IMU- > ttyCH343USB0。

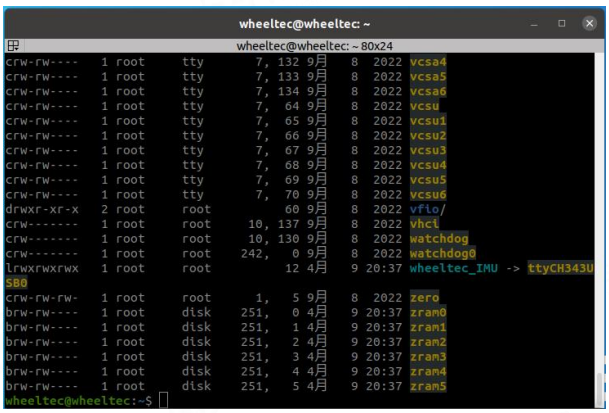


Figure 5.2.3 Device alias final effect

As a result, the reality has successfully used the alias wheeltec_IMU to represent the inertial navigation module, and no matter which USB port is connected, it does not need to consider the port number change problem when using it.

5. 3 ROS_SDK Function Pack

Copy the files under the yesense_imu.zip compressed files in the [WHEELTEC H30 inertial navigation data\2. ROS_SDK] folder to the src folder in the workspace and decompress them.

Open the terminal, run the following command in the workspace path to compile, and wait for the [100%] compilation success is displayed:

```
catkin_make -DCATKIN_WHITELIST_PACKAGES=yesense_imu
```

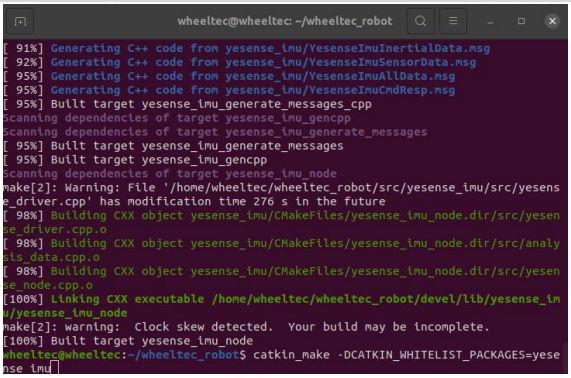


Figure 5.3.1 Compile the SDK

Enter the command to run to turn on inertial navigation

```
roslaunch  yesense_imu  yesense_ahrs.launch
```

Enter the command to print the data of the IMU

```
rostopic echo /imu
```

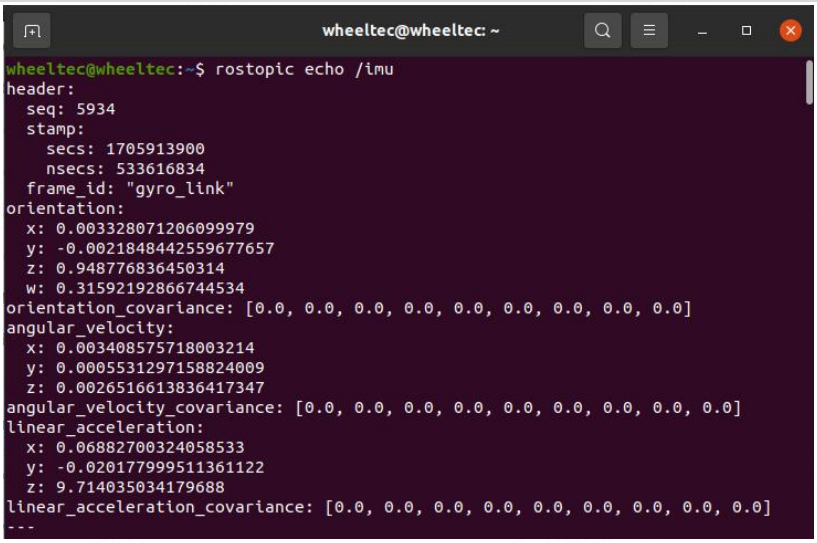


Figure 5.3.2 Data viewing interface

Use of IMU visualization:

Before starting the IMU module (yesense_ahrs.launch), if you need to verify the pose of the IMU module, you can set the [is_visual] parameter to "true" and re-run the command to turn on inertial navigation

```
roslaunch yesense_imu yesense_ahrs.launch
```

```
yesense_ahrs.launch
<launch>
  <!-- IMU_TF is visualization ? -->
  <arg name="is_visual" default="false"/>
  <node pkg="yesense_imu" type="yesense_imu_node" name="yesense_imu_node" required="true" output = "screen" >
    <!-- 串口设备, 可通过rules.d配置固定-->
    <param name="yesense_port"      type="string" value="/dev/wheeltec_IMU_NE1"/>
      <!-- 波特率 -->
    <param name="yesense_baudrate"  type="int"    value="460800"/>
    <!-- 发布的imu话题中的frame_id -->
    <param name="frame_id"  type="string"    value="gyro_link"/>
    <!-- 发布的imu话题名 -->
    <param name="imu_topic"  value="imu"/>
  </node>
   <include file="$(find yesense_imu)/launch/tf.launch" if="$(arg is_visual)"/>
  <!--<node pkg="rviz" type="rviz" name="rviz" args="-d $(find yesense_imu)/rviz/yesense.rviz" required="true"/>-->


</launch>

```

Enter the command rviz in the terminal and load the [yesense_imu/rviz/yesense.rviz] file configuration, and you can see the visual pose of the IMU, as shown in the following figure:

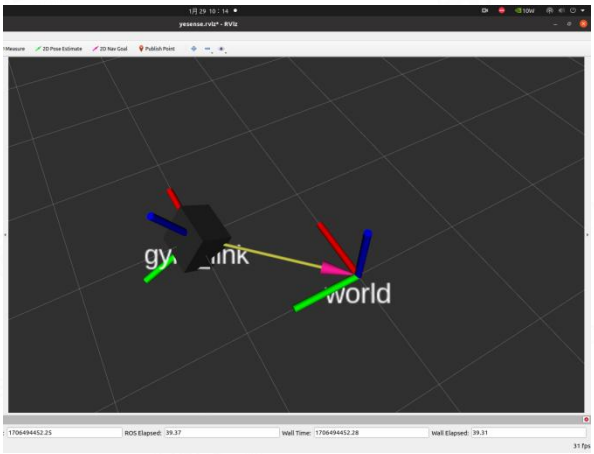


Figure 5.3.3 IMU visualization pose

5. 4 ROS2_SDK Function Pack

Copy the files in the compressed yesense_ros2.zip file in the [WHEELTEC H30 Inertial Navigation Data \2.ROS_SDK] folder to the src folder of the workspace and

decompress the file.

Open the terminal, run the following command in the workspace path to compile, first need to compile the data interface related files:

```
colcon build --packages-select yesense_interface
```

After compiling successfully, enter the following instructions in the terminal to compile the source code file:

```
colcon build --packages-select yesense_std_ros2
```

If an error occurs during compilation and serial's library is missing, the driver for serial_ros2 needs to be reinstalled.

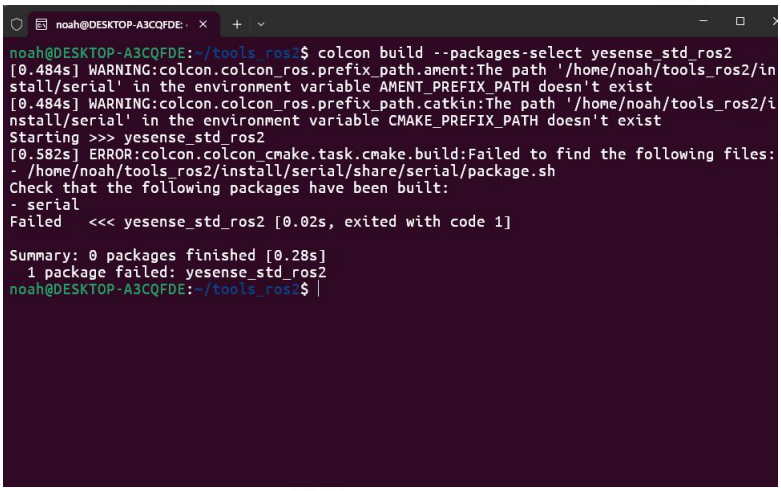


Figure 5.4.1 The compiling SDK error is missing serial

After compiling, connect the IMU module to the computer and enter the command to open the Inertial Navigation

```
ros2 launch yesense_std_ros2 yesense_node.launch.py
```

Enter the command to print the imu data

```
ros2 topic echo /imu_data_ros
```

6. Mechanical dimensions

6. 1 Description of dimensions

WHEELTEC H30 product dimensions are shown below.

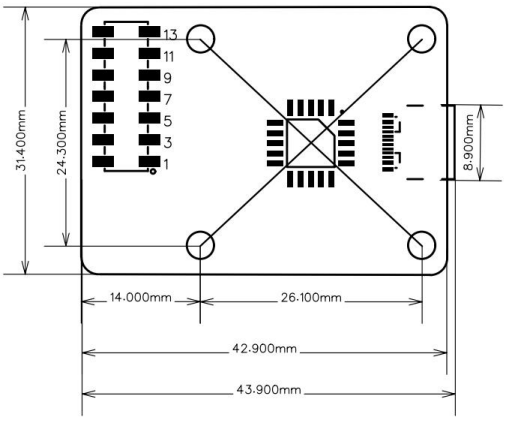


Figure 6.1.1 WHEELTEC H30 Board size

6. 2 Packing description

The WHEELTEC H30 module is encapsulated in an anti-static seal bag.

7. Summary of frequently questions

7. 1 Module output not normal

Q：The module has no data output.

A:

1. Use the serial port assistant to check whether the serial port of the module has output.
2. If the serial port has output but the host computer has no data, change the baud rate. You can find the packet output in “59 53” and change the baud rate of the host computer to the corresponding baud rate.
3. If the serial assistant has no output, you need to use the serial assistant to send the command to set the output content and see whether there is a return value.
4. If no value is returned in the three tests, replace the hardware (cable, DK, etc.) and try again.
5. If you try to replace the hardware without success, please contact our after-sales (during the warranty period).

Q：The module has output, but failed to configure using the host computer or instructions.

A：The baud rate of the host computer may not match the baud rate of the module; The baud rate is too small, and the output content is too much; The serial port communication is blocked.

1. Check whether the output data of the host computer is normal (baud rate matching).
2. Try to reduce the frequency, or reduce the output content of the configuration is successful (serial communication).
3. Check whether increasing baud rate can be successful (if the baud rate is too small, the configuration fails).
4. If the communication is faulty, replace the cable or DK board, or power on the device again. If the baud rate is incompatible with the output content, refer to the formula in the manual to calculate the available baud rate.

7. 2 Heading angle drift

Q：The heading angle of the module output has been drifting (it has been changing when stationary).

A:Generally, it is caused by the excessive zero bias value of the gyroscope, and the zero bias calibration can be carried out by the host computer or instructions.

Q：When the module uses the AHRS algorithm mode, the heading angle is temporarily drifted or inaccurate.

A：Generally, due to the interference of the surrounding magnetic field, or there is no magnetic calibration, only the magnetic calibration can be carried out through the host computer.

7. 3 IIC communication without data

Q：When using IIC communication, the data cannot be read

A:When acquiring data for IIC communication, it is necessary to wait for the state of the INT pin level to be pulled up before the data can be obtained through the IIC, and the INT pin has a short pull time each time, and an external interrupt needs to be set to capture the level status. If the INT pin is not pulled up, the data cannot be obtained.

7. 4 Problems that may arise in practical applications

1. If the algorithm mode of the module is AHRS, the module will be calibrated with magnetism after the random motion stops, and the heading angle will converge, which is normal.
2. If the module is a non-AHRS module, the attitude of the module in random motion is calculated according to the calculation, there will always be errors, when the module motion is stationary, the module will use gravitational acceleration to converge the attitude, then there will be a change in the attitude angle.
3. If the movement is too vigorous and exceeds the range of the module, it will also cause the attitude of the module to slowly converge.
4. The customer rotates along the Z axis of the module, theoretically the pitch angle and roll angle will not change, but because the horizontal plane of the customer's rotation is different from the horizontal plane of our calibration turntable, it will cause some small changes in the roll angle and pitch angle, which is inevitable.